Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “system” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Bring back an earlier idea

Two carts approach: one has momentum 5 kg·m/s east and the other 3 kg·m/s west.
Explain why adding their magnitudes gives the wrong total; find the signed total.

Word help: momentum = mass times velocity, including direction

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$.
Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

Your turn: Draw a tip-to-tail momentum diagram and label each axis with units.

Sentence starter: The … in my model represents … . Its label shows … .

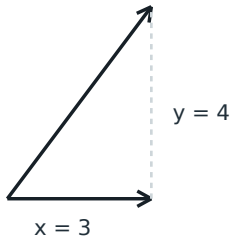
Marzano check - circle: ① ② ③

03 Look and explain

Two incoming arrows point east and north.
Sketch the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: The … in my model represents … . Its label shows … .



The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

Momentum in two dimensions

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

Incoming total momentum is 6 kg·m/s east and 8 kg·m/s north. Two objects stick with total mass 2 kg. Divide each component by 2 to find velocity components.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Use the idea

Why must both momentum component sums use the same system and same time interval?

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

06 Choose a method

An ideal asteroid orbit has nearest distance 4 AU and farthest distance 8 AU. Where is it fastest?
Explain with the equal-area law.

Word help: orbit = a path around another body under gravity

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

07 Exit check

Total momentum is 4 kg·m/s west and 2 kg·m/s south. Combined mass 2 kg.
Give final velocity components.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A joined pair of total mass 2 kg has total momentum 6 kg·m/s east and 8 kg·m/s north.
Find its velocity components using each momentum component divided by 2.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3